

AI Voice Assistant Platform – Technical & Commercial Feasibility Proposal

Executive Summary

This proposal describes a custom multilingual AI voice platform for healthcare service providers. The client already owns a virtual number, allowing integration with the existing telephony provider rather than purchasing new numbers. The platform will answer inbound calls, make outbound calls, qualify leads, schedule callbacks, analyze conversations, generate dashboards, and integrate with CRM systems.

Recommended Architecture

Customer → Existing Virtual Number → Telephony Provider (or Exotel SIP/Webhook) → FastAPI Backend → Speech-to-Text → GPT-4o/Claude → Business Logic → Text-to-Speech → Customer. The backend stores transcripts, recordings, lead details, callback schedules and analytics in PostgreSQL.

Core Features

Multilingual (English/Hindi/Gujarati/Marathi), AI FAQ handling, service discovery, lead qualification, callback scheduling, appointment booking, CRM integration, WhatsApp/SMS notifications, dashboards, sentiment analysis, service-wise analytics, recordings and transcripts.

Development Plan

Phase 1: Requirements (1 week) Phase 2: Telephony integration (2 weeks) Phase 3: AI conversation engine (3 weeks) Phase 4: Dashboard & analytics (3 weeks) Phase 5: CRM/calendar integrations (2 weeks) Phase 6: UAT & deployment (2 weeks) Estimated MVP: 10–13 weeks.

Technology Stack

FastAPI, React, PostgreSQL, Redis, OpenAI GPT-4o (or Claude), Deepgram (STT), ElevenLabs/Cartesia (TTS), Existing Telephony or Exotel, Docker, AWS/Azure.

Infrastructure Cost

Approximate infrastructure cost per connected minute: Telephony: ₹0.60–2.00
Speech-to-Text: ₹0.20–0.70 LLM: ₹0.40–2.00 Text-to-Speech: ₹0.30–1.20
Hosting/Storage/Monitoring: ₹0.30–0.90 Typical total: ₹3–5 per connected minute.

Commercial Pricing Recommendation

One-time implementation: ₹8–12 lakh (MVP) Monthly platform/support: ₹25,000–60,000 Usage: ₹6–10 per connected minute. This leaves margin above infrastructure while covering maintenance and support.

Competitor Comparison

Managed AI platforms: Ringgg AI, Retell AI, Vapi, Bland AI, Synthflow. Telephony providers: Exotel, Knowlarity, Airtel IQ, Twilio. Custom platform advantages: ownership of data, healthcare workflows, custom dashboards, white-labeling, flexible integrations.

ROI

Reduced missed calls, automated lead qualification, 24x7 availability, service analytics, reduced manual workload, faster follow-ups, improved conversion rates.

Conclusion

A custom platform is commercially viable when the client requires customization, branding, ownership, and integration with existing systems. Managed AI platforms are suitable for rapid pilots, while a custom build provides long-term control and recurring SaaS revenue.